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DSA0402
Fundamentals of Data Science

CO2 - AT 1

1 A csv file is imported into pandas, but the first row of actual data is wrongly treated as column names. What is the best correction?

A Use header = None while reading the file.

2. A dataset has 25 columns, but only age, salary and department are required for analysis. Why should only these columns be selected?

A To reduce unnecessary data processing.

3 A manger wants employees from the IT department with salary above ₹ 60,000 which command is correct?

A df[df['Department'] == 'IT' ] &(df['Salary'] > 60000)]

4 A student dataset is filtered using: df[df['Marks'] >=50]
what does this operation return?

A Student who scored 50 or above.

5 A dataset has 40% missing values in one column. What is the most analytical concern before filling missing values?

A Wheater missingness may affect the reliability of analysis.

6 Mean manipulation is most suitable when:

A The numerical data is approximately symmetric

7 A Salary column contains a few very high executive salaries. Some salaries values are missing. which component

A Median.

8 A city column contains missing values. which method is usually suitable?

A Replace missing values with the most frequent city.

9 A dataset has 10,000 rows and only 12 rows contain missing values. what is a reasonable approach?

A Drop the 12 incomplete rows.

10 A company wants to identify the top 5 highest paid employees. which sequence is most appropriate?

A Sort salary in decending order and select first 5 rows.

11 A school wants to sort students first by class and then by marks in descending order within each class. which Command is correct?

A df.sort_values(['Class', 'Marks'], ascending = [True, False])

12 A Sales dataset contains Region, Product, and Sales. To find total sales for each region and product combination, which Operation is best

A df.groupby(['Region', 'Product'])['Sales'].sum()

13 A college wants to compare average marks department-wise which result best supports this analysis?

A Mean marks grouped by department.

14 A Company wants department wise average salary and maximise salary. which command is suitable?

A df.groupby('Department')['Salary'].agg(['mean', 'max'])

14 Two students have the same mark. What does ranking help identify?

A It assigns unique positions while handling ties based on rank method.

15 A company wants to rank employees by salary within each department. Which command is correct?

A df['Rank'] = df.groupby('Department')['Salary'].rank(ascending=False)

16 A recruitment wants to compare average marks of five departments which visualization is best suitable?

A Bar Chart

17 A company records monthly profit for 24 months. which chart gives the best analytical understanding of growth or decline?

A Line chart.

18 A dataset contains Advertising Budget and Sales Revenue. which visualization is most suitable to analyze whether higher advertisement spending increase Sales?

A Scatter plot.

19 A dataset contains employee salaries, and the analyst wants to know extremely high salaries which visualization is most useful?

A Box Plot.